

Math 520, F07
Quiz 3

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. Given the following differential equation:

$$t + ye^{2ty} + (y^2 + te^{2ty}) \frac{dy}{dt} = 0$$

(a) Show that this equation is exact.

Proof: $M(t, y) = t + ye^{2ty}$, $N(t, y) = y^2 + te^{2ty}$
 $\frac{\partial M}{\partial y} = e^{2ty} + 2tye^{2ty}$
 $\frac{\partial N}{\partial t} = e^{2ty} + 2tye^{2ty} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$
Thus, this equation is exact.

(b) Find the general solution of this equation.

Solution: By (a), we know there exists a function $\phi(t, y)$ such that
(i) $\frac{\partial \phi}{\partial t} = M(t, y)$ and (ii) $\frac{\partial \phi}{\partial y} = N(t, y)$.

From (i), we get

$$\phi(t, y) = \int (t + ye^{2ty}) dt + h(y) = \frac{t^2}{2} + \frac{e^{2ty}}{2} + h(y)$$

From (ii), we get

$$\phi(t, y) = \int (y^2 + te^{2ty}) dy + k(t) = \frac{y^3}{3} + \frac{e^{2ty}}{2} + k(t)$$

$$\text{So that } h(y) = \frac{y^3}{3}, k(t) = \frac{t^2}{2}.$$

$$\text{Thus, } \phi(t, y) = \frac{t^2}{2} + \frac{e^{2ty}}{2} + \frac{y^3}{3} = C \text{ is the general solution.}$$

(c) Find the specific solution of this equation with initial condition $y(0) = 3$.

Solution: Put $t=0$, $y=3$ into the general solution obtained in (b), we get

$$C = \frac{0^2}{2} + \frac{e^0}{2} + \frac{3^3}{3} = \frac{1}{2} + 9 = \frac{19}{2}$$

$$\text{So } \frac{t^2}{2} + \frac{e^{2ty}}{2} + \frac{y^3}{3} = \frac{19}{2} \text{ is the specific solution.}$$